**33** A metal wire has a length of 2.50 m and a cross-sectional area of $4.50 \times 10^{-6}\,m^2$. The resistivity of the metal is $3.50 \times 10^{-7}\,\Omega m$.

The wire is stretched so that its length increases to 2.65 m. The wire remains cylindrical and the **volume** of the wire remains constant.

What is the change in the resistance of the wire?

**A** $0.012\,\Omega$ **B** $0.024\,\Omega$ **C** $0.19\,\Omega$ **D** $0.22\,\Omega$